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LECTURER'S NAME: DANIEL DOBRE

STUDENT'S NAME: MOSES LOGAH AMPOFO

STUDENT'S INDEX NUMBER: PU/BIT/2025/007

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Question 1

INTRODUCTION

Organizations rely more on information technology to operate efficiently, coordinate activities, and respond to competitive pressures (Laudon & Laudon, 2022). The report discusses how information systems function and how hardware and software applications support business activities. It first examines the fundamentals of information systems and then discusses business information systems and their main types.

Major Components of Information Systems

An information system comprises interrelated components that are coordinated to collect, process, store, and distribute information. This process aids in coordination, control, analysis, and decision-making within an organization (Laudon & Laudon, 2022). Further, it includes a feedback loop which allows the performance to be monitored and improved (Stair and Reynolds, 2018).

There are five major components of Information Systems (O'Brien & Marakas, 2011). The first is hardware, or the actual machines that communicate and process data, as well as the devices used to capture and store data and to present it, including computers, servers, scanners, and mobile devices. The second is software, which comprises programs that direct hardware operations. Application software like accounting packages and customer relationship management packages carry out specific functions for a business, while system software such as an operating system controls computing resources. The third one is data, which is the raw data that the system collects and stores, typically in databases. The fourth is people, the people who develop, operate, and use the system: the end users, analysts, programmers, and managers. The fifth is procedures, the formal

rules and policies that govern the use of systems, such as data entry standards, security controls and backup routines. These are all linked: Technology can't provide value without competent users and clear procedures.

Data and Information

Data are unprocessed facts, they do not have meaning of their own. Information, on the other hand, is the data that has been processed and is useful for the recipient (Laudon & Laudon, 2022). Thus, data is fed into a system, and information is generated out of it. Information is usually presented in the form of summaries, reports and trends that are directly related to decision making, while data are normally presented as single figures or records.
The difference is highlighted by two practical examples of business. Data is gathered from the individual point-of-sale records of items sold, quantities, prices, and times in a retail environment. The information these records provide can be used by management to change buying and staffing when they are analyzed weekly to give them information about the best-selling products and the busiest trading days. In human resources, an employee's daily clock-in and clock-out times are data. A monthly summary of hours worked, late arrivals, and overtime for each department is information, because it feeds into payroll and performance reviews.

Major Functions of an Information System

Information systems perform four principal functions (O'Brien & Marakas, 2011; Stair & Reynolds, 2018). The first is input, in which the system captures data from internal and external sources through devices such as keyboards, barcode scanners, and online forms. Data quality at

this stage is critical because errors introduced here propagate through later stages. The second is processing; this is where data is converted into information through calculation, sorting, classification, and summarization. For example, when payroll is computed or sales are aggregated by product. The third is storage, in which data and information are retained in databases or cloud platforms for future retrieval, supporting record keeping, auditing, and historical analysis. The fourth involves output and communication, disseminating information to users via reports, dashboards, invoices or alerts and distribution through networks to ensure the right people are informed of the decision. This helps to increase the flow of information in the organization and with customers and suppliers. Feedback and control are present throughout all four functions, enabling the system to identify an error, assess performance, and adjust (Stair & Reynolds, 2018).

Question 2

BUSINESS INFORMATION SYSTEMS

Business information system (BIS) is an information system that has been developed to serve various business functions, including its operation, management, and business strategy (O'Brien & Marakas, 2011). It provides functional support for business functions such as sales and marketing, finance and accounting, manufacturing, and human resources, as well as for the operational, managerial and strategic levels of business organization (Laudon & Laudon, 2022). Business information systems can enhance the quality of decision making, the efficiency of business processes, and help achieve a competitive advantage in the business. Business information systems come in various forms and several types.

- **Transaction Processing Systems (TPS):** TPS carry out and enter the standardized transactions required for everyday business operations like Order Entry, Payroll, Reservations, etc. (Laudon & Laudon, 2022). It is defined by high volumes of structured, repeated processing requiring speed, accuracy and reliability. Other systems follow TPS since it is their primary source of data, and if TPS is poor, other systems are poor. Examples consist of point-of-sale systems, ATM's and payroll processing.
- **Management Information Systems (MIS):** MIS gather and organize data obtained from the TPS to form summary and exception reports that are used by middle level managers to monitor, control and plan operations (Laudon & Laudon, 2022). Much of its work is repetitive and formulaic, likely to compare actual performance with the targets or the budget. Reports such as monthly sales report, inventory status and budget-to-actual reports are examples of these types of reports.

- **Decision Support Systems (DSS):** A Decision Support System (DSS) is an interactive, computer-based system that enables users to gain a better understanding of the problems they face and construct analytical models to facilitate the decisions they need to make in dealing with semi-structured and unstructured problems (Sprague, 1980). Primarily, it is flexible – this allows the user to carry out "what if" analyses and evaluate alternatives before investing resources. Common examples involve financial forecasting models, delivery route optimization and credit risk evaluation.

- **Executive Information Systems (EIS) and Executive Support Systems (ESS):** An EIS which became ESS, is a single, easily accessible means of delivering to senior executives consolidated and aggregated internal and external information relevant to strategy (Rockart & Treacy, 1982). These systems generally feature graphic interfaces and include "drill-down" functions that enable executives to go from general indicators to specific subtleties. They can aid long term planning based on key performance indicators, market trends and competitor data.

CONCLUSION AND RECOMMENDATIONS

Information system is combination of hardware, software, data, network, people, and procedure which converts raw data to information through the input, processing, storage and output. Business information systems utilize this function organization-wide: TPS collect transactional data, MIS aggregate or summarize transactions for control purposes, DSS provide analysis for complex problems and EIS/ESS provide information for analyzing strategy.

It is strongly suggested that management first improve the transaction processing systems as the reliability of all the systems above depends on the accuracy of the transaction data. The resultant MIS reporting should then be rolled out to enhance the monitoring and control facilitated by middle management and thereafter, decision support tools and executive dashboards should be incrementally introduced as the needs of the organization evolve and its capabilities develop. Investments in staff training, data security, and properly defined procedures are crucial at each step, as the positive impact of technology is reliant on the individual users and the processes they follow (Laudon & Laudon, 2022).

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